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REVIEW



Understanding the role of vegetation root systems in the initiation of rainfall-induced shallow landslides: scaling perspectives

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ABSTRACT

The contribution of root systems, defined as the arrangement of belowground structures formed within plant communities through both individual plant root systems and their interactions, to the stability of forested hillslopes can be assessed at individual root and plant levels. Root systems influence both vertical and lateral flow by creating preferential flow pathways or bypassing flow and influencing pore-water development and slope stability. Root systems development differs depending on stand density (i.e. tree spacing) and species composition, which are also associated with resource competition among individual plants. Forest stand conditions related to plant density and resultant root system development possibly alter frequency and size of landslides. Scaling effect, defined as translation of insights of individual root system into entire hillslopes or watersheds, poses challenges due to the heterogeneity of forest stand conditions, soil properties, topography, and hydrological processes. The primary goal of this review is to explore how forest root systems and forest stand conditions influence landslide dynamics through hydrological and mechanical processes, thereby providing insights for further research to address sustainable forest management and disaster risk reduction.

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effects

Introduction

Vegetation root systems, refer to the composite belowground structures formed by plant communities, encompassing both individual plant root systems and their interactions within ecosystems. Vegetation root systems of trees, shrubs, and herbaceous plants alter the stability of forested hillslopes (Tsukamoto and Kubota 1991; Sidle 1992; Freschet et al. 2021). The development of vertical and lateral roots systems characterizes both hydrological and geotechnical conditions in soil matrix (Ghestem et al. 2011; Zhang et al. 2020). Vertical roots, which grow deep into the soil, act at the basal shear surface (Abe and Ziemer 1991), enhancing shear resistance (Schmidt et al. 2001; Cohen and Schwarz 2017). Lateral roots, which grow parallel to the hillslope (Tsukamoto and Kubota 1991; Freschet et al. 2021), form physical connections with the root systems of neighboring plants (Stokes et al. 2009), thereby increasing resistance against slope instability.

Root architecture, defined as the spatial arrangement and structures of plant roots, including their root diameters, distribution, density, and branching, influences slope stability (e.g. Lynch 1995). Depending on root architecture, root pull-out resistance, which is the force required to pull a root out of the soil, and root bending strength, which is the root's resistance to bending forces, alter anchoring forces of soil and influence slope stability (Abe and Iwamoto 1986). In particular, Jiang et al. (2023) demonstrated that deep-rooted species such as elms (*Ulmus procera*) increase slope stability by enhancing the shear strength of soil. Kinoshita et al. (2013) revealed the importance of lateral root strength for slope stability in both coniferous and broadleaf forest. Löbmann et al. (2020) highlighted the role of lateral roots

from grass species (e.g. *Anthoxanthum odoratum* and *Trifolium pratense*) in slope stabilization, particularly in protecting against shallow landslides by forming a dense root mat that binds the surface soil, reduces erosion, and enhances slope stability.

Various methods have been developed for evaluating the effects of vegetation root systems on slope stability based on field measurements, laboratory experiments, and modeling. By examining soil sample, soil with root can exhibit a 20–40% increase in shear strength compared to rootless ones (e.g. Wu et al. 1988; Yildiz et al. 2018). Sato et al. (2009) demonstrated that tree roots contribute to slope stability by reinforcing the soil and increasing the safety factor by 0.04 to 0.28 based on measurement of pull-out resistance of the roots ranged from 0.5 to 2.3 kN/m². Estimated slope stability by using factor-of-safety also increased from 1.0 to 1.5 depending on root cohesion (e.g. Ziemer 1981; Flepp et al. 2021).

Spatial extent of root systems such as distribution and connectivity of neighboring roots needs to be included in evaluating root strength and hillslope stability (e.g. Sakals and Sidle 2004; Schwarz et al. 2010). Sakals and Sidle (2004) showed that the root size of Douglas fir (*Pseudotsuga menziesii*) varies with the distance from the tree stem, based on field investigations and a root inventory conducted to examine slope stability in the Oregon Coast Range (U.S.A.). Based on the correlation between diameter at breast height (DBH) and the effective radius of the rooting zone, Roering et al. (2003) found that the effects of roots on soil reinforcement are significant in conifer-dominated forests of the Oregon Coast Range. Despite the progress in upscaling from individual trees to hillslope scales, root demography is typically

obtained at the individual plant level (distance between stem and quantity of roots: e.g. Genet et al. 2008; Mehtab et al. 2021) and does not consider interactions among individual roots.

For a more comprehensive understanding of slope stability in forested landscapes, the interactions of roots among neighboring plants need to be incorporated. Root interactions occur through various ecological processes, including growth and competition during long-term vegetation succession (e.g. Yamase et al. 2024), which optimize plant growth, maximize soil stabilization, and ensure effective nutrient and water uptake (Wang, Zhao et al. 2020; Francini et al. 2021). Plant species have varied root architectures, such as taproots, fibrous roots, and adventitious roots, which influence their ability to stabilize slopes (Mickovski and van Beek 2009). Lateral extent of the rooting zone also differs depending on species and stand density (Johnsen et al. 2005). Furthermore, some plants possess root system development for avoiding competition, others are more tolerant of close spacing (Mattia et al. 2005; Yamase et al. 2024). Soil texture and structure influence root system development; for instance, sandy soils with high porosity facilitate root penetration (Greenway 1987). High fertility and water availability in soils can support closer plant spacing because nutrients are more readily available, whereas poor soils require wider spacing to reduce competition (Gray and Sotir 1996).

Despite substantial advances in understanding the role of vegetation root systems in hillslope stability, a significant knowledge gap remains when integrating ecological interactions related to root system development at the hillslope scale. Changes in root system development, related to ecological conditions, also alter the physical properties of soil and the internal hydrological processes within the soil matrix (Ghestem et al. 2011; Wang, Ma et al. 2020). Some studies suggest that root systems positively influence slope stability by acting as preferential flow pathways that drain water (e.g. McGuire et al. 2016); others report negative effects because roots can facilitate the accumulation of pore-water pressure on hillslopes (e.g. Li et al. 2021). This study synthesizes existing knowledge regarding the effects of vegetation root systems on slope stability, including the physical properties and hydrological functions of roots. We also demonstrate the importance of a scaling perspective from individual- to hillslope-scale dynamics in relation to root system development and discuss potential factors that control landslide initiation at the hillslope scale.

Physical properties of root systems affecting slope stability

Contribution of root systems on slope stability is considered as root cohesion (ΔC), which refers to the additional shear strength conferred to the soil by the presence of roots within the soil matrix (Sidle and Ochiai 2006). The modified Mohr–Coulomb failure criterion incorporates root cohesion into the soil shear strength equation (Schmidt et al. 2001). Slope stability is then evaluated by the ratio of resisting forces to traction forces. The overall soil shear strength (S : kN/m^2) is expressed as:

$$S = C' + \Delta C + (\sigma - u) \tan \phi \quad [1]$$

where ΔC is root cohesion (kN/m^2), C' is the effective soil cohesion (kN/m^2), σ (kN/m^2) is the normal stress by overburden pressure consisting of the weight of soil column and additional external load, ϕ is the internal friction angle of soil ($^\circ$), and u is pore-water pressure (kN/m^2) calculated using height of water column (m) and unit weight of water (kN/m^3).

Root cohesion (ΔC) is estimated based on the strength of individual root (Figure 1), determined by tensile or pull-out tests (e.g. Sidle 1992; De Baets et al. 2008; Fan et al. 2021), while also considering root distribution (Figure 1). Root bundle models (RBMs) are then used to estimate root cohesion of a group of roots. The first bundle model, developed by Wu et al. (1979) and Waldron (1977), known as the Wu and Waldron model (WWM), assumes that root reinforcement can be represented by summing the individual strengths of different roots, reflecting an increase in shear strength as the number of roots increases based on the following equation,

$$\Delta C = k \sum_{i=1}^N T_{r,i} \times RAR_i \quad [2]$$

where $T_{r,i}$ is the root strength (kN/m^2) for the i^{th} root diameter class, RAR_i is the root area ratio or the root biomass concentration for the i^{th} root diameter class, and k is a factor related to root geometry. RAR is defined as the fraction of total root area within soil cross-sectional area (e.g. Gray and Leiser 1982). The WWM assumes that root tensile strength is fully functioned when the soil–root composite material fails, meaning all roots break simultaneously. Hence, the estimation of root reinforcement using Equation [2] tends to overestimate root reinforcement by up to 300% (Pollen and Simon 2005).

Revised RBM was proposed based on the observation of progressively failures of roots by variations of strength and elasticity with respect to diameter (Schwarz et al. 2012) (Figure 1). Fiber bundle model (FBM) developed by Pollen and Simon (2005) accounts gradual breaking of roots based on breaking threshold. Because shear force is redistributed among the remaining intact roots, overall reinforcement contribution is reduced depending on remaining root cohesions. This is expressed as:

$$\Delta C = k \times \max(T_{r,i} \times RAR_i \times i) \quad [3]$$

where i is the number of weakest roots that are still intact upon loading the root bundle of N roots, RAR_i is the RAR of root i , and $T_{r,i}$ is the strength of the weakest intact root, $\max$ refers to the maximum value of the product for a given set of roots which represents the maximum cohesion by the intact roots (Pollen and Simon 2005). The RBM extends FBM through incorporating the strain-step loading approach, which incrementally applies small strain (displacement) steps to the root bundle to account for progressive failure of individual roots as the displacement (strain) increases (Schwarz et al. 2011). In the RBM, root strength ($T_{r,i}$) is calculated as a force contribution from each root or root class, incorporates variability through the Weibull survival function (Schwarz et al. 2013). The Weibull survival function defines the

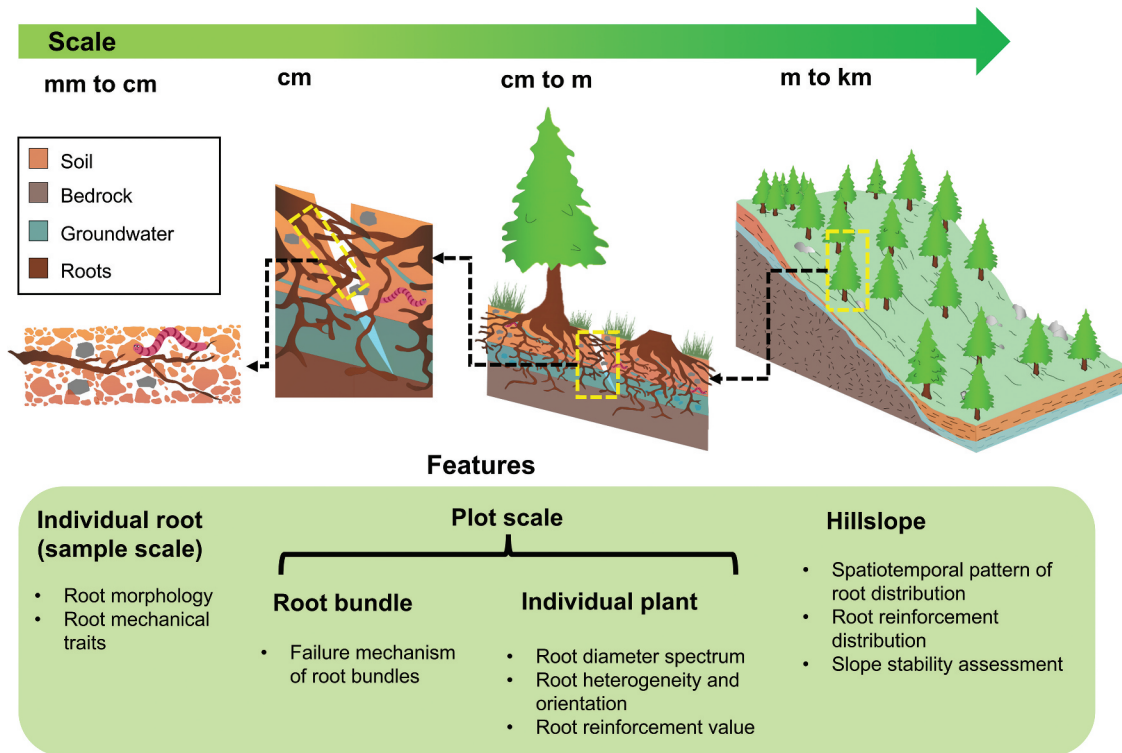


Figure 1. Schematic illustration of the scale of root reinforcement measurements, spanning multiple spatial dimensions. Measurements at the millimeter to centimeter (mm to cm) scale capture fine root dynamics, root–soil interface interactions, and tensile strength related to root morphology and mechanical traits. At the centimeter scale, studies focus on root bundles related to root architecture and branching patterns at plot scale. The centimeter to meter (cm to m) scale incorporates the structural reinforcement provided by individual plant roots and their influence on soil stability over localized areas. At the meter to kilometer (m to km) scale, root systems are evaluated within the context of forest stands and landscapes, examining their collective role in mitigating landslides and hydrological processes. This multi-scale approach highlights the interconnected contributions of roots to soil stabilization across varying spatial extents.

probability of a root surviving beyond a specific displacement, using a power–law relationship where the survival probability decreases exponentially with increasing displacement (Genet et al. 2005). However, Pollen (2007) noted that despite these improvements, FBMs maintain limited predictive accuracy due to their inability to consider the dependency of root tensile strength on soil hydrological processes, a crucial factor in triggering rainfall-induced shallow landslides. Furthermore, Ji et al. (2020) demonstrated that discrepancies between landslide prediction using FBM and observed root cohesion may be associated with pull-out of root (not breakages of root but slipping roots from soil matrix) which is not considered in any model development.

To assess the stability of vegetated hillslopes, the modified Mohr–Coulomb failure criterion (Equation 1) is applied in slope stability analysis using limit equilibrium approaches (e.g. Fellenius method, Bishop’s simplified method, and Janbu’s method), which divide the slope into slices and analyze the equilibrium of forces acting on each slice (e.g. McGuire et al. 2016). Finite element methods, such as the strength reduction method, progressively reducing the shear strength parameters of soil or rock until failure occurs, Kokutse model, a two-dimensional finite element model simulating the mechanical effects of plant root systems and tree surcharge, and Ecosfix 1.0, a three-dimensional slope stability model designed to evaluate the mechanical stability of forested hillslopes under various forest management scenarios considering horizontal and vertical spatial distributions of soil and root properties, also are used to simulate

slope behavior, incorporating soil and root properties (Mao et al. 2014).

Hydrological processes for the interaction of root systems and slope stability

Hydrological processes such as rainfall infiltration, soil water content dynamics, and the development of soil saturation can alter tensile strength of root systems and ultimately relates to slope stability (Lann et al. 2024) (Figure 2; Table 1). Fan et al. (2021) showed that changes in water content within the soil matrix reduced tensile strength in pull-out tests from 200 to 50 kPa as the saturation level increased from 30% to 90% based on field investigations of paper mulberry (*Broussonetia papyrifera*) in China. Schwarz et al. (2010) revealed that pull-out became 30% low when the volumetric water content elevated from 0% to 10%, based on laboratory pull-out tests. Shuin et al. (1998) showed that high soil moisture condition (low soil suction) reduced soil cohesion using a laboratory experiment with bamboo sticks. These results suggested that distribution of water content in the soil matrix affects the level of reinforcement by root cohesion and the resultant slope stability (Hales and Miniati 2017).

The presence of roots in the soil matrix also influences hydrological processes at both pedon and hillslope scales (Sidle et al. 2017). At the soil sample scale, roots enhance conductivity primarily by creating macropores and altering soil aggregation (Figure 2; Table 1). As roots grow and penetrate into soil, roots form pores that increase transmissivity of water by up to 10 times (Angers and Caron 1998).

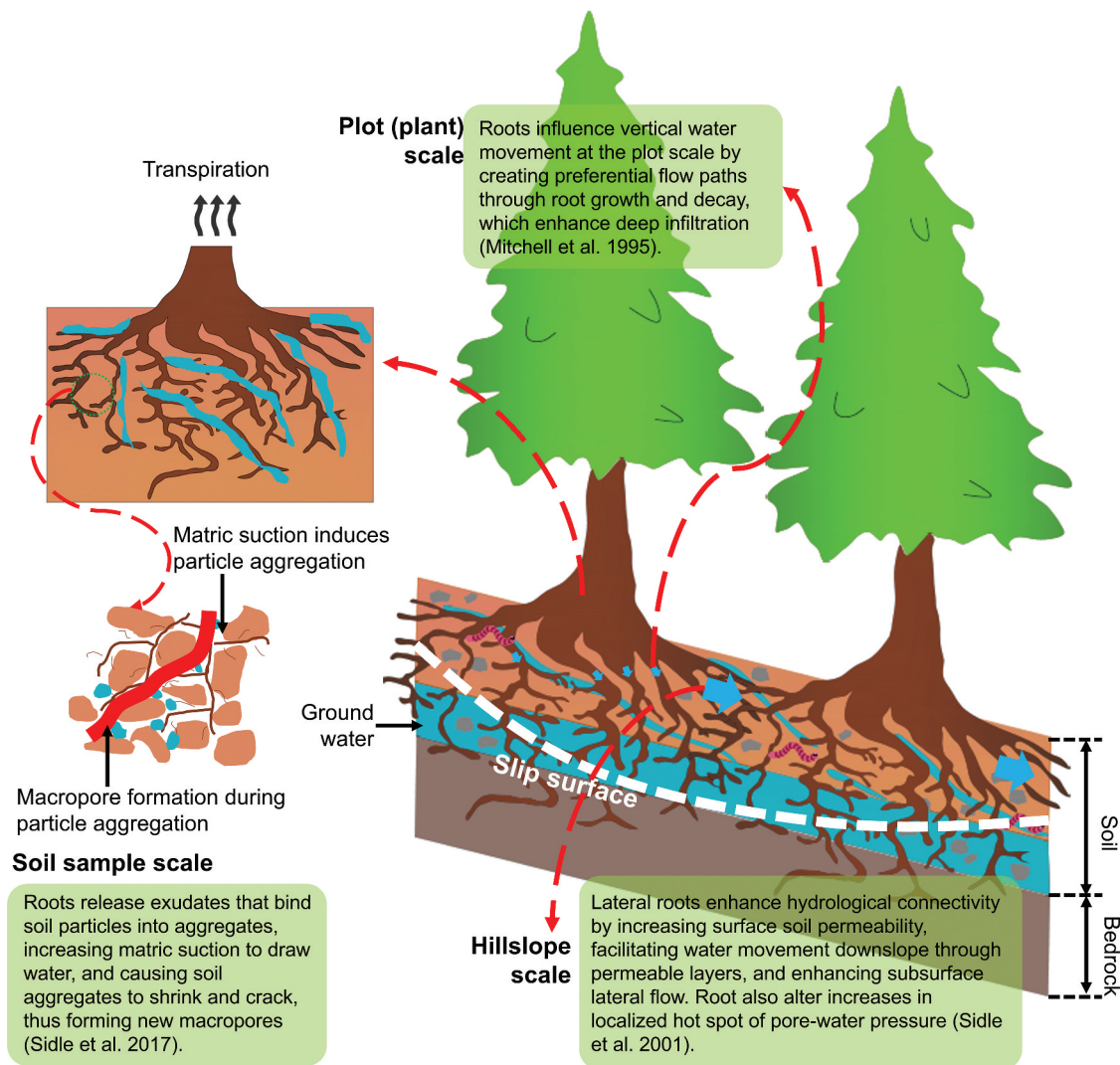


Figure 2. Hydrologic processes affected by root systems at different scales. At the soil sample scale (centimeters to decimeters), roots improve soil hydraulic properties such as porosity and saturated hydraulic conductivity by creating micro- and macropores. These pores improve water infiltration and retention. At the plot scale (meters), individual tree roots induce preferential flow through both living and dead roots. Localized saturation may occur depending on the concentration. Roots also extract water for transpiration, resulting in localized dry zones. At the slope scale (tens to hundreds of meters), root networks collectively reinforce the soil, reducing surface runoff and promoting vertical and lateral water movement. Roots slow subsurface flow by enhancing root systems and hydraulic connectivity, mitigating rapid water transport during storms, and delaying streamflow and subsequent saturation. Across scales, root systems are critical for regulating water dynamics and maintaining slope stability.

Root exudates, which are organic compounds released by roots, contribute to soil aggregation by bonding smaller particles (0.5–5 mm) into aggregates larger than 5 mm (Baumert et al. 2018; Kang et al. 2023), creating a well-structured soil with interconnected pore spaces that facilitate water movement (Bronick and Lal 2005). Consequently, the presence of root systems can enhance the hydraulic conductivity of soil by up to 6 times, from 7.2 to 43.2 mm/h, depending on root architecture, such as tap-like and heart root systems (Wang et al. 2020). Leung et al. (2015) found significant influence of roots on soil water retention, while plant transpiration is a recognized driver of changes in soil pore water pressure. Increased suction by transpiration draws more water toward the root zone from wetter areas, enhancing the soil's ability to supply water to plants during dry conditions (Raats 2007). Changes in matric suction can also cause soil aggregates to shrink and crack, leading to the formation of new macropores or the expansion (both size and length) of existing ones (Assouline 2006; Sidle et al. 2017).

Root systems influence vertical water movement through preferential flow paths in soil matrix (Figure 2). Preferential flow path is developed by roots grow and decay (Ghestem et al. 2011), creating complex channels from the soil surface to deeper layers. Collected water from branches and leaves via stem flow can be delivered along the roots: this process is conceptualized as double -funneling (Johnson and Lehmann 2006). During rainfall events, water can move through these flow path, bypassing the soil matrix (Jarvis 2007) and altering the distribution of soil water content (Liang et al. 2011). Wang et al. (2007) showed that the roots (150 cm deep) of desert shrubs (*Artemisia ordosica* and *Caragana korshinskii*) create preferential flow pathways that enhance infiltration to a depth of 60 cm during extreme rainfall events, increasing soil-water storage for prolonged dry periods. Extensive root channels in the hillslope of a Dahurian larch (*Larix gmelinii*) forest in northeast China enhance infiltration rates by up to 700% and reduce surface runoff, contributing to improved soil moisture retention and reduced erosion risks (Wang et al. 2022).

Lateral roots contribute to hydrological connectivity across different spatial scales (e.g. Noguchi et al. 1999;

Table 1. Experimental studies evaluating the contribution of roots to slope dynamics in chronological order.

No	Root system analog		Experimental method	Key issues	Key words	Reference
	Actual	Artificial				
1	<i>L. leucocephala</i>		Laboratory experiment	Biodiversity and biomass	Shear strength, pioneer species	Osman and Barakbah (2011)
2	Multiple species		Field experiment	Slope gradient and vegetation cover	Erosion, runoff	El Kateb et al. (2013)
3	Farmland, grassland, and forest		Field experiment	Forest vs. grassland	Runoff, sediment erosion	Zhang et al. (2014)
4		Cellulose acetate	Centrifuge simulation	Root shape and pull-out resistance	Soil friction, root architecture	Kamchoom et al. (2014)
5		Acrylonitrile butadiene styrene	Centrifuge simulation	Earthquake and stability	Woody roots, numerical modeling	Liang et al. (2015)
6		Cellulose acetate	Centrifuge simulation	Root architecture and stability	Heart-shaped roots, shear strength	Leung et al. (2017)
7	Multiple species		Field experiment	Local plant species for re-vegetation	Local species, mechanical stability	Ranjan et al., (2016)
8		Polyester fibers	Centrifuge simulation	Vetiver grass roots and infiltration	Groundwater, shear strength	Likitlersuang et al. (2017)
9		Acrylonitrile butadiene styrene	Centrifuge simulation	Seismic stability and vegetation	Vegetation, slope morphology	Liang and Knappett (2017a)
10		Acrylonitrile butadiene styrene	Centrifuge simulation	Prediction of landslides by earthquake	Vegetation, numerical model	Liang and Knappett (2017)
11	<i>Cynodon dactylon</i>		Field experiment	Pyrenees slope stability	Soil moisture, rainfall pattern	Oorthuis et al. (2018)
12	<i>Cynodon dactylon</i>		Field experiment	Gravitational erosion in cold regions	Freeze–thaw, Vegetation seeding	Rui et al. (2018)
13	<i>Phalaris arundinacea</i>		Field experiment	Plant species and stability in trophic	Soil stability, soil chemical content	Demenois et al. (2018)
14	<i>Cynodon dactylon</i>		Field experiment	Probabilistic slope stability	Cohesion, frictional angle	Das et al. (2018)
15	<i>Melastoma malabathricum</i> and <i>Bauhinia purpurea</i>		Field experiment	Mixed vegetation effectiveness	Soil stability, shrubs, trees	Li et al. (2018)
16	Grass, deciduous, coniferous forest		Field experiment	Coniferous trees and soil water	Suction, seasonal change	Hayati et al. (2018)
17	<i>Schefflera Arboricola</i>		Field and laboratory experiment	Root spacing and stability	Root cohesion, tensile strength	Ni et al. (2019)
18		Acrylonitrile butadiene styrene	Centrifuge simulation	Root model comparison	Cohesion, slip planes	Liang et al. (2019)
19	<i>F. suspensa</i> , <i>K. japonica</i>		Laboratory experiment	Shrubs and soil erosion:	Landslide, gully formation	Gao et al. (2020)
20		Polyurethane solid rubber rod	Laboratory experiment	Tree planting and slope angle:	Dense planting, management	Lan et al. (2020)
21	<i>Ligustrum x vicaryi</i>		Laboratory experiment	Root water uptake	Suction, water-holding capacity	Meng et al. (2021)
22	<i>Nephrolepis cordifolia</i> , <i>Zoysia japonica</i>		Laboratory experiment	Species comparison	Slope stability, water storage	Chen et al. (2021)
23		Polylactic acid plastic	Laboratory experiment	Root volume and plant spacing	Surface cracks, soil deformation	Ding et al. (2022)

Gomi et al. 2008) and resultant soil water distribution (Hales and Miniati 2017) (Figure 2). The development of lateral roots enhances infiltration in the surface soil horizon by up to 144% (Shi et al. 2021), because more than 90% of lateral roots are located within the top 60 cm of the soil (Coutts and Lewis 1983; Jackson et al. 1997). Ide et al. (2010) showed that significant amount of preferential flow occurred near soil surface due to high lateral root density of bamboo. Similarly, in Japanese cypress forest, Sidle et al. (2001) suggested that the presence of shallow lateral roots enhances subsurface lateral flow by creating more permeable pathways for water movement near soil surface.

The effect of root systems on subsurface lateral flow depends on their architecture. For example, downward-sloping lateral roots are particularly effective at draining water (Ghestem et al. 2011). Forked roots can divide

water flow, whereas converging forks downslope may increase or decrease water pressure in the surrounding soil matrix (Gish et al. 1998; Hao et al. 2021). Sinuous roots can also affect water flow because soil matrix containing curved roots becomes a hotspot for pore-water pressure development (Ghestem et al. 2011) (Figures 2 and 3). The ends of both living and dead roots can serve as discontinuous flow paths, leading to localized pore-water pressure development (Ghestem et al. 2011; Shao et al. 2017). Interconnected root channels improve water drainage, whereas isolated root channels are more likely to induce pore-water pressure development (Sidle et al. 2001; Ghestem et al. 2011).

Root system, overlapping, and landslide characteristics

The root system and their overlapping enhance the overall

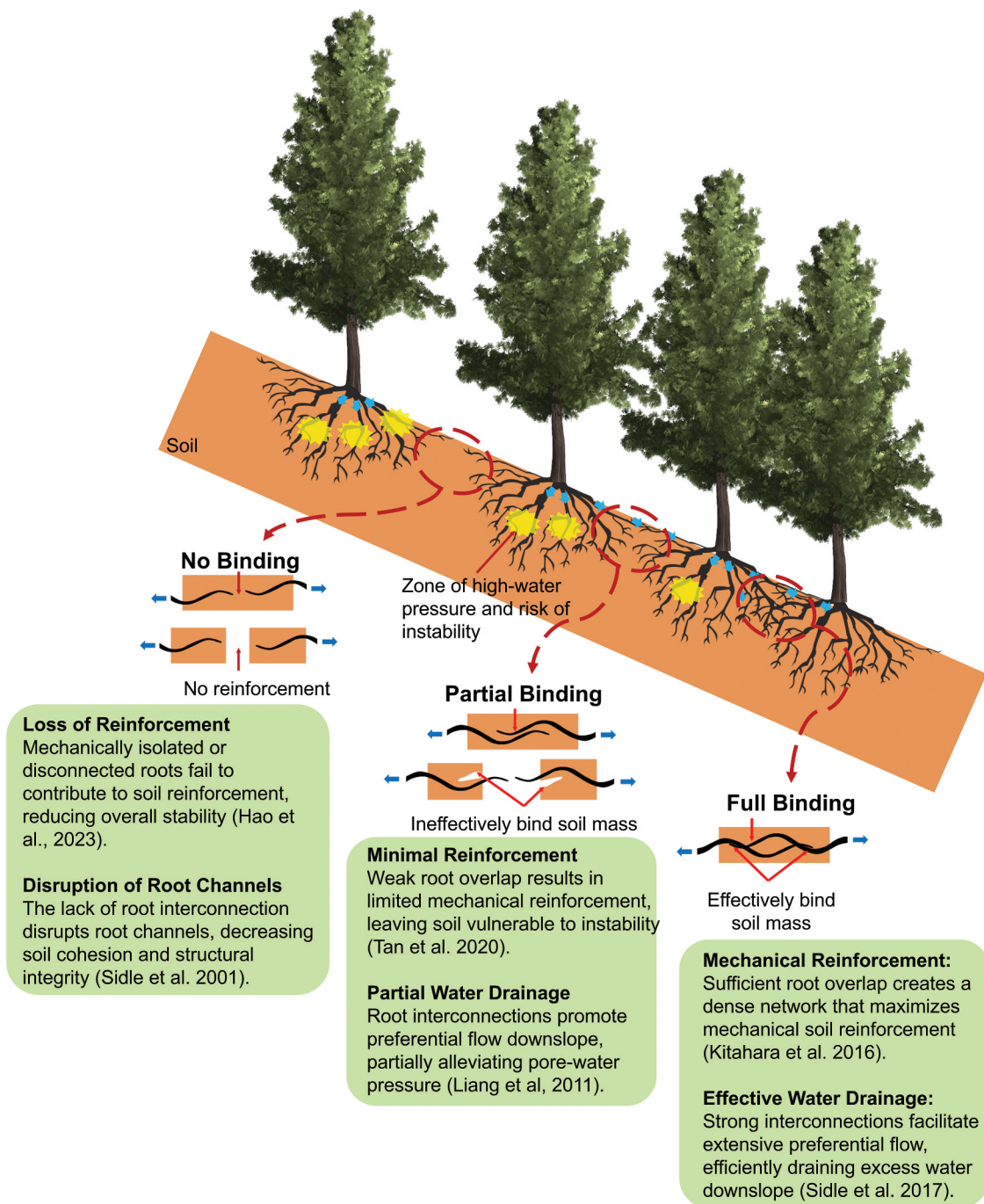


Figure 3. Effects of root interconnection on the mechanical and hydrological functions of root systems. No binding: disconnected roots provide no soil reinforcement, leading to instability. Poor drainage causes pore-water pressure buildup, gap for the root systems possible for slope failure risk. Partial binding: some reinforcement and drainage occur, but uneven connectivity allows localized pore-water pressure buildup, maintaining moderate instability. Full binding: dense root networks maximize soil reinforcement and drainage, ensuring slope stability. For instance, root resistance increases up to 15 kN/m in dense root systems as per Schwarz et al. (2016).

contribution of vegetation to slope stability (e.g. Sakals and Sidle 2004; Schwarz et al. 2010) (Figure 3). Mechanically isolated roots can create vulnerable zones with the potential to initiate slope failure (Tsukamoto et al. 1986; Schwarz et al. 2012; Mao et al. 2014) (Figure 3). For instance, Yamase et al. (2021) found that tree gaps are potential hotspots for landslide initiation, even in the presence of root overlap. Cohen and Schwarz (2017) revealed that spaces between trees are potential locations for landslide initiation during rainstorms, even under high tree density based on numerical simulation. On the other hands, sufficient root overlap creates a dense

root network that maximizes mechanical soil reinforcement. Strong interconnections of root system facilitate extensive preferential flow, efficiently draining excess water downslope, and developing saturated zones in the lower part of hillslope (Figure 3).

Root system development and their overlapping also related to landslide size because lateral root overlap amplifies resistance at the boundaries of a potential landslide and reduces the spatial extent of landslide (Milledge et al. 2014). Based on observations after the 2018 Eastern Iburi earthquake in Japan, Ritonga et al. (2021) found that

landslide size was smaller in secondary deciduous and conifer forests than that in logged areas due to extensive lateral roots and their reinforcement. Furthermore, using three-dimensional limit equilibrium analysis, Schwarz et al. (2016) revealed that lateral root cohesion of 15 kN/m in a Poplar Forest (*Populus × euramericana*) with a density of 2500 stems/ha on non-cohesive soils led to a 65% reduction in the size of rainfall-induced landslides. Lan et al. (2020) showed that the smallest landslide area occurred in hillslope with highest vegetation density (100 plants/m² at 10 cm spacing) that that in hillslope with lower density.

The presence of root systems may not always lead to a reduction in landslide size. For instance, Rickli and Graf (2009) found larger landslide areas on forested hillslopes than that on non-forested ones in a field study of rainfall-induced landslides in Switzerland. Li et al. (2021) revealed that increasing lateral root cohesion from 30 to 100 kPa doubled the scar length of landslides. Deng et al. (2022) demonstrated that the mean size of landslides in the Central Loess Plateau of China was 50% greater in areas with the highest tree density (NDVI 0.64–0.8 vs. <0.62). High lateral root cohesion between soil particles or soil blocks leads to the formation of large soil masses (i.e. large soil blocks) within the soil layers, resulting in extensive landslide areas (Bellugi et al. 2015). The conflicting relationship between vegetation density and changes in landslide area is likely influenced by root interconnection (Stokes et al. 2009), the formation of soil blocks by roots, and the breakage of root connections during the early stages of landslides (Hales and Miniati 2017). Furthermore, a high degree of root interconnection may influence the formation of localized pore-water pressures, contributing to soil instability on hillslopes (Figure 3).

Effects of stand density and species composition on root systems

The development of root systems is related to the vegetation density and interactions between neighboring plants (Bormann and Gordon 1984) (Figure 4). Dense vegetation typically results in stronger physical connections between adjacent root systems (Norris et al. 2008). For example, numerical modeling of Norway spruce (*Picea abies*) showed that a tree density of 4,400 stems/ha achieved twice the maximum root cohesion compared to a density of 1,600 stems/ha (Cohen and Schwarz 2017). Similarly, a 34% increase in root cohesion was associated with increases in tree density from 100 to 400 stems/ha in ash (*Fraxinus excelsior*) forest, resulting expanded network of interconnected roots in overlapping zones (Schwarz et al. 2012).

The effect of stand density on slope stability is linked to root cohesion (shear strength), which diminishes with increasing distance from the tree's center. Near the tree's base, primary roots, with ≥10 cm of diameters, provide substantial mechanical reinforcement to soil. In contrast, at greater distances, fine lateral roots dominate, offering significantly less tensile strength, often measured at less than 1 MPa (Nicoll and Ray 1996). Exponential decreases of root cohesion (ΔC : kN/m²) can be expressed as:

$$\Delta C = aD^b \quad [4]$$

where D is the distance from the tree center (m), and a and b are coefficients. For Japanese cypress (*Chamaecyparis obtuse*) stand, Kitahara et al. (2016) suggested that b is approximately -2.8 . Kitahara et al. (2016) also proposed that coefficient a is related to DBH (cm) based on the following,

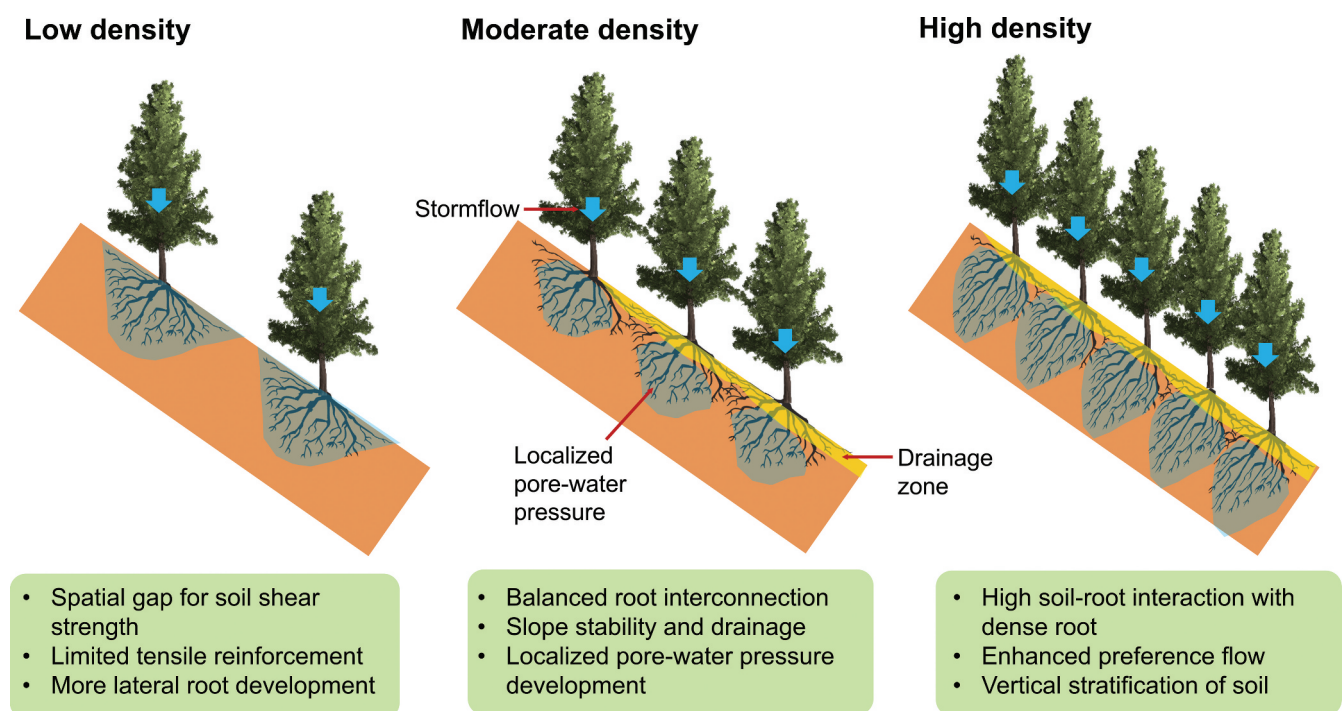


Figure 4. Effects of stand density on root interconnection and the resultant mechanical and hydrological functions of root systems. Low density: root systems are sparsely distributed, resulting in weak root interconnection and fewer roots bind soil particles. Infiltration via root systems is limited. Moderate density: sufficient lateral and vertical spread of root occurs with soil cohesion and mechanical strength. Localized pore-water pressure by preferential flow occurs. High density: dense root systems induced lateral flow at vertically stratified soil. Localized pore-water pressure may buildup. Higher stand densities (e.g. >2500 trees/ha) significantly reduce pore-water pressure but increase gravitational forces based on empirical studies (e.g. Lan et al. 2020).

$$a = 0.0033DBH^{2.77} \quad [5]$$

Using the Equations [4] and Equation [5], root cohesion ΔC can be represented solely by the distance from the tree center and the tree's DBH. This reduction in root tensile strength with increasing distance highlight the critical role of tree density (spacing) and root architecture in slope stability. Kitahara et al. (2016) also proposed equations for estimating root cohesion with distance of tree center for various tree species.

The physical connection of root systems through fusion and grafting possibly enhance hillslope stability (Figure 4). For instance, Graham and Bormann (1966) found that American elms (*Ulmus americana*) frequently form natural root grafts with neighboring trees. Tarroux et al. (2010) observed natural root grafts in white pine (*Pinus strobus*) under conditions of shared resources and stress conditions. In a field experiment in Estonia, Külla and Lõhmus (1999) showed that Norway spruce (*Picea abies* (L.) Karst.) possesses roots capable of fusing and grafting, fostering root interconnections among individuals of the same species, depending on stand and rooting densities. This grafting occurs when the roots of neighboring trees come into contact and fuse, forming a connection (Graham and Bormann 1966). Such connections enable trees to share nutrients, water, and even chemical signals (Stokes et al. 2009). Natural root graft has various implications, such as facilitating communication and resource sharing between trees, as well as the potential for disease or pathogen transmission (Graham and Bormann 1966). Moreover, the interlocking of roots with rocks and coarse fragments in the soil enhances the lateral membrane strength of the soil–root complex in stands with certain densities.

Effects of vegetation density on slope stability was also examined by considering the effects of thinning (density control) related to forest management (Figure 4). Kakeya et al. (2016) examined how thinning practices increase root volumes of 10- to 30-year-old stands by comparing thinned and un-thinned Japanese cedar (*Cryptomeria japonica*) plantation forests. Sato et al. (2001) revealed that trees in a forest subjected to 9-year thinning (600 stems/ha) had greater vertical and lateral root volume compared to non-thinning sites (2075 stems/ha). On the other hand, Yamase et al. (2019) showed that root strength estimated by RBM between thinned (723 stems/ha) and un-thinned (1625 trees/ha) did not significantly differ. Changes in stand density through forest management influence lateral root growth and the spatial distribution of root cohesion around tree trunks (Kitahara et al. 2016).

Root systems development relates to the competition among individual plants for nutrients and water (Schenk 2006; Yamase et al. 2024). In a study of *Eucalyptus globulus* trees in Western Australia, Coomes and Grubb (2000) found that competition for resources drives roots to penetrate deeper into the soil, which is less exploited by neighboring plants. Based on findings from the global root database, Tomobe et al. (2023) suggested that competition for space restricts lateral root spread, leading to more vertical growth than horizontal spread in deserts, semi-deserts, scrublands, grasslands, and shrub- and tree-savannas. Herbaceous plants such as Mediterranean salt-bush (*Atriplex halimus*) and Spanish thyme (*Thymus zygis*) tend to develop extensive lateral root systems that spread

horizontally near the surface (De Baets et al. 2008), allowing them to exploit resources in the topsoil (Löbmann et al. 2020). These extensive root systems intertwine with neighboring root systems, creating a surface biomat that effectively drains water (Gerke et al. 2015; Löbmann et al. 2020).

The presence of multiple species intensifies interspecific competition, promoting root system development and increasing interactions among individual plants. Root system diversity can be defined by the coexistence of various root architectures, including taproots, fibrous roots, and shallow lateral roots (Tsukamoto et al. 1982; Jackson et al. 1997). Landslides that initiate from mixed-stand forests differ from those in monocultures. Schmidt et al. (2001) reported that the total landslide volume in a natural forest consisting of mixed Douglas fir and red cedar was up to seven times smaller than that in a monoculture red alder (*Alnus rubra*) forest in the Oregon Coast Range. Bakker et al. (2021) demonstrated that deep-rooted species in mixed-species grass plots performed better when surrounded by shallow-rooted neighbors. In an 8-year experiment, belowground biomass increased by up to 28% as plant species richness expanded from one to sixteen species (Ravenek et al. 2014). The study also indicated that combining plant functional groups, such as those with nitrogen-fixing abilities or high photosynthetic efficiency, significantly influenced the diversity of root distribution. An experimental study in the Bailong River Basin (China) demonstrated that the proportions of soil aggregates in a natural hillslope dominated by mixed grasses and shrubs versus a monoculture black locust (*Robinia pseudoacacia*) stand were 43.2% and 23.0%, respectively, indicating that mixed species exert a greater influence on soil structure aggregation than monoculture plantations (Wang et al. 2019). While many studies have investigated the impact of stand density and composition on root system development (e.g. rooting depth and biomass), their influence on landslide initiation remains largely unexamined.

Scaling from a single plant to the hillslope

Scaling up from individual plant root systems to hillslope and catchment scales is crucial for assessing slope stability and understanding landslide dynamics in forested landscapes (e.g. Fu et al. 2016; Masi et al. 2021) (Figure 1). This complexity of hillslope scale associated with internal hydrological processes, soil development, and stand density (also species composition) (Figure 2). Various hillslope hydrology such as water movement, development saturation, and soil-water retention affect root systems development (Ghestem et al. 2011). Forest stand condition (e.g. density and species composition) adds another layer of complexity scaling (Figure 4). For instance, Loades et al. (2010) discovered that higher plant density generally enhanced slope stability, although resource competition resulted in smaller interconnected root diameters. Likewise, Ni et al. (2019) found through field experiments that small-diameter roots (<1 mm) exhibited lower tensile strength at a vegetation spacing of 0.5 m compared to 1.1 m. Furthermore, the presence of forest gaps larger than 20 m increases landslide susceptibility due to reduced root strength within the gap (Moos et al. 2016). Besides forest structure, densely spaced trees can significantly add weight to hillslopes, heightening

the risk of slope movement, especially in steep or already unstable regions (Lotfalian et al. 2019).

The conventional calculation of soil reinforcement associated with root cohesion on a hillslope scale typically aggregates the contributions of adjacent individual roots or root bundles (Sakals and Sidle 2004; Schwarz et al. 2010). This approach assumes that physical connections between root systems exhibit mechanical behavior similar to individual roots. However, the resistance provided by such connections may differ from that of individual roots, depending on the extent of hydrological and ecological processes. For instance, hydrological processes involving dead and live tree roots can influence preferential flow (bypass flow) (Sidle et al. 2001; Liang et al. 2011), leading to localized saturations at hillslopes and alterations in hillslope instability (Figure 2). In ecological processes, herbaceous species frequently establish extensive root interconnection as their roots spread and intertwine with neighboring plants (Löbmann et al. 2020). Conversely, woody species may encounter challenges in establishing such extensive interconnection due to their propensity to avoid competition and the limited space available (Stokes et al. 2009). These findings highlight the need for comprehensive slope stability assessments that include both hydrological and ecological processes to fully understand the complex interactions between soil and vegetation.

Numerous laboratory and field experiments have been conducted for capturing the functions of roots on slope stability using scaled artificial and real roots (Table 1). Experimental approaches control vegetation density and species compositions for capturing the effects of vegetation on soil-water interactions (e.g. Ding et al. 2022). Some field-based experiment was conducted for testing soil-water responses at mixed vegetation conditions and slope stability (e.g. Gk et al. 2018; Li et al. 2018); however, eliminating site-specific effects remains challenging (Ochiai et al. 2007). Small-scale experiments provide valuable insights into the relationship between vegetation root systems and slope stability (Liang et al. 2015) (Table 1). Hence, replication in experiments is essential for ensuring reliability and reproducibility, as it helps validate results and reduce the impact of random variability. Experiment including replications is often challenging due to variability and constraints related to vegetation and soil interaction as well as large-scale equipment and costs.

Small-scale experiments can be useful in dealing with problems related to replication. However, this approach has had various issues, such as how gravity affects soil mass and how to model vegetation roots. Because gravitational force acting on mass is small in small scale experiments compared to the actual landslides (Kessler et al. 2018), root cohesions in the small flume experiments are possibly overestimate. Furthermore, because initial motion of landslide is important for the occurrence of landslides (Iverson 2015), different gravitational force acting on soil mass between small scale experiment and feed may affect the function of roots (tensile strength and breakages) and subsequent mobility of landslide mass. For overcoming these problems, Kessler et al. (2018) proposed applying

various size of scale experiments for examining landslide initiation and mobility. Additionally, numerous site-specific factors in field settings make it difficult to isolate the effect of vegetation alone. 3-dimensional root structures using Structure from Motion (SfM) and 3D printing is also effective approaches for including root architectures in the experiment (Liang and Knappett 2017b). Hence, this model does not include the biological complexity and interconnection which observed in living roots. Despite recent progress in landslide experiments with plants, we still do not understand how the vegetation conditions (e.g. density) affect the ecological, water, and soil processes on hillslopes that lead to landslides (Table 1).

Future challenges

Addressing challenges related to root systems and slope stability requires a comprehensive approach that integrates research methodologies with various scales. One important challenge lies in understanding the complex interaction of roots, soil, and water. Water dynamically influences the mechanical strength of both soil and root systems (e.g. Fan et al. 2021), often leading to reduced slope stability during periods of high moisture content (e.g. Hales and Miniati 2017). To mitigate this issue, a combination of enhanced measurement techniques (e.g. Yamase et al. 2018), monitoring (e.g. Arata et al. 2020), experimental studies (e.g. Noviandi et al. 2022), and advanced simulations (e.g. Ng et al. 2021) is necessary. Ground-penetrating radar can non-destructively assess tree root biomass, aiding in the understanding of root distribution and its role in slope stability (Yamase et al. 2018). The implementation of detailed hydrological monitoring systems, such as soil moisture sensors and piezometers, can provide real-time data regarding water content in slopes, allowing early detection of conditions that could lead to landslides (Arata et al. 2020). Controlled laboratory experiments simulating different hydrological scenarios and root system development can deepen our understanding of how moisture levels affect reinforcement by root cohesion as well as timing and magnitude of landslides.

Another major challenge is understanding root architecture, which is crucial for anchoring soil and enhancing slope stability. Root architecture greatly varies among species and is influenced by factors such as stand density and hydrogeomorphology (Schenk and Jackson 2002; Eshel and Beekman 2013). However, knowledge in this area is limited due to the complexity of root systems and the variability in their development. To address this knowledge gap, a comprehensive database cataloging the root architectures (e.g. rooting depth, branching, and root distribution radius) of various species in different settings should be developed. One of the unique approaches was conducted by Makoto et al. (2023) for analyzing the relationships between 227 tree species of specific leaf area and maximum rooting depth. Such analysis with large-scale meta-analyses helps to identify common patterns and correlations that could inform predictive models above ground conditions and below-ground root system development. In addition, studies focusing on a specific type of tree need to explore how different types of roots develop in response to different levels of trees, nutrients, topography,

and types of trees along hillslope (Yanagisawa and Fujita 1999). Evaluating the impact of tree root architectures on slope stability in steep terrains is crucial for understanding and mitigating landslide risks. For instance, root development varies with slope inclination, with some species exhibiting higher root density downslope and others upslope; considering these differences is essential for effective forest management and slope stabilization strategies (Tsutsumi et al. 2004). This knowledge is vital for improving forest management efforts aimed at enhancing slope stability.

Scaling issues present a major challenge in landslide assessment at landscape scales. Multiple forest types and forest stand density need to be examined for evaluating slope stability at large scales. Although previous studies focused on stand ages and root development for slope stability (Larcher 2003; Imaizumi et al. 2008; Genet et al. 2008), stand density (also tree spacing) alters the lateral root connections and overlapping. Forest stand density also vary depending on the management legacies (commercial thinning and/or forest fire control). Forest succession also may affect root strength by transitioning from shallow, dense root systems in early successional stages to deeper, stronger root system connection in mature forests, enhancing long-term soil reinforcement by root cohesion (Cammeraat et al. 2005). Species composition changes and root decay during succession can create temporary gaps in stability, but the establishment of late-successional species restores and strengthens slope resilience. Additionally, succession influences hydrological dynamics (Cornish and Vertessy 2001), such as soil moisture and rainfall interception, further impacting root effectiveness and overall slope stability. For identify changes in stand condition by management and succession stages, technologies such as air borne LiDAR, remote sensing, and terrestrial laser scanning can be effective to examine forest stand condition such as stand density (Takahashi et al. 2010). Comprehensive database of forest inventory that include information on stand density, species composition, and root architecture will enable findings to be upscaled from small-scale studies to larger regions, ultimately improving landslide mitigation strategies on a broader scale. Such information can be integrated as root-soil-water interaction model for simulating landslide occurrence in forested landscape.

Conclusion

This review highlights the critical role of root systems in slope stability by synthesizing findings from studies on vegetation, root strength, and landslide mitigation. Through an integrative analysis, we identified consistent patterns and mechanisms by which root systems enhance soil cohesion, reduce shear stress, and stabilize slopes under varying environmental conditions. The review consolidates evidence from field measurements, laboratory experiments, and modeling studies, providing a comprehensive understanding of root-system functions across different forest types and landscapes. Scaling is crucial for understanding the role of root systems in hillslope stability. These issues greatly vary across scales, from micro-scale interactions between roots and soil particles to macro-scale influences involving interactions within a vegetation community. At the hillslope scale, stand density and the resulting physical connections between neighboring root systems regulate soil stability and landslide risk through

ecological, geotechnical, and hydrological mechanisms. The dynamic nature of forests, from establishment through succession (characterized by changes in stand density), highlights the importance of forest management practices that balance resource competition with root physical connections. The effects of stand density, root interconnection, and associated ecological, geotechnical, and hydrological processes on hillslope stability require further exploration to improve forest management practices. This review also underscores the need for collaborative, cross-disciplinary efforts to develop sustainable forest management strategies that capitalize on the protective functions of root systems in vulnerable landscapes. Overall, implementing density-based forest management and fostering a diverse, resilient, interconnected root systems are crucial for efforts to reduce landslide risks and promote sustainable land use in forested landscapes.

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References

- Abe K, Iwamoto M. 1986. An evaluation of tree-root effect on slope stability by tree-root strength. *J Jpn For Soc.* 68(12):505–510. doi: [10.11519/jjfs1953.68.12_505](https://doi.org/10.11519/jjfs1953.68.12_505).
- Abe K, Ziemer RR. 1991. Effect of tree roots on a shear zone: modeling reinforced shear stress. *Can J For Res.* 21(7):1012–1019. doi: [10.1139/x91-139](https://doi.org/10.1139/x91-139).
- Angers DA, Caron J. 1998. Plant-induced changes in soil structure: processes and feedbacks. *Biogeochemistry.* 42(1/2):55–72. doi: [10.1023/A:1005944025343](https://doi.org/10.1023/A:1005944025343).
- Arata Y, Gomi T, Sidle RC. 2020. Topographic features and stratified soil characteristics of a hillslope with fissures formed by the 2016 Kumamoto earthquake. *Geoderma.* 376:114547. doi: [10.1016/j.geoderma.2020.114547](https://doi.org/10.1016/j.geoderma.2020.114547).
- Assouline S. 2006. Modeling the relationship between soil bulk density and the water retention curve. *Vadose Zone J.* 5(2):554. doi: [10.2136/vzj2005.0083](https://doi.org/10.2136/vzj2005.0083).
- Bakker LM, Barry KE, Mommer L, Van Ruijven J. 2021. Focusing on individual plants to understand community scale biodiversity effects:

- the case of root distribution in grasslands. *Oikos*. 130 (11):1954–1966. doi: [10.1111/oik.08113](https://doi.org/10.1111/oik.08113).
- Baumert VL, Vasilyeva NA, Vladimirov AA, Meier IC, Kögel-Knabner I, Mueller CW. 2018. Root exudates induce soil macroaggregation facilitated by fungi in subsoil. *Front Environ Sci*. 6:422975. doi: [10.3389/fenvs.2018.00140](https://doi.org/10.3389/fenvs.2018.00140).
- Bellugi D, Milledge DG, Dietrich WE, Perron JT, McKean J. 2015. Predicting shallow landslide size and location across a natural landscape: application of a spectral clustering search algorithm. *J Geophys Res Earth Surf*. 120(12):2552–2585. doi: [10.1002/2015jf003520](https://doi.org/10.1002/2015jf003520).
- Bormann BT, Gordon JC. 1984. Stand density effects in young red alder plantations: productivity, photosynthate partitioning, and nitrogen fixation. *Ecology*. 65(2):394–402. doi: [10.2307/1941402](https://doi.org/10.2307/1941402).
- Bronick CJ, Lal R. 2005. Soil structure and management: a review. *Geoderma*. 124(1–2):3–22. doi: [10.1016/j.geoderma.2004.03.005](https://doi.org/10.1016/j.geoderma.2004.03.005).
- Cammeraat E, van Beek R, Kooijman A. 2005. Vegetation succession and its consequences for slope stability in SE Spain. *Plant Soil*. 278:135–147.
- Chen J, Lei X, Zhang H, Lin Z, Wang H, Hu W. 2021. Laboratory model test study of the hydrological effect on granite residual soil slopes considering different vegetation types. *Sci Rep*. 11(1):14668. doi: [10.1038/s41598-021-94276-4](https://doi.org/10.1038/s41598-021-94276-4).
- Cohen D, Schwarz M. 2017. Tree-root control of shallow landslides. *Earth Surf Dyn*. 5(3):451–477. doi: [10.5194/esurf-5-451-2017](https://doi.org/10.5194/esurf-5-451-2017).
- Coomes DA, Grubb PJ. 2000. Impacts of root competition in forests and woodlands: a theoretical framework and review of experiments. *Ecol Monogr*. 70(2):171–207. doi: [10.1890/0012-9615](https://doi.org/10.1890/0012-9615).
- Cornish PM, Vertessy RA. 2001. Forest age-induced changes in evapotranspiration and water yield in a eucalypt forest. *J Hydrol*. 242 (1–2):43–63. doi: [10.1016/S0022-1694\(00\)00384-X](https://doi.org/10.1016/S0022-1694(00)00384-X).
- Coutts MP, Lewis GJ. 1983. When is the structural root system determined in Sitka spruce? *Plant Soil*. 71(1–3):155–160. doi: [10.1007/BF02182650](https://doi.org/10.1007/BF02182650).
- De Baets S, Poesen J, Reubens B, Wemans K, De Baerdemaeker J, Muys B. 2008. Root tensile strength and root distribution of typical Mediterranean plant species and their contribution to soil shear strength. *Plant Soil*. 305(1–2):207–226.
- Demenois J, Carriconde F, Bonaventure P, Maeght J, Stokes A, Rey F. 2018. Impact of plant root functional traits and associated mycorrhizas on the aggregate stability of a tropical ferrallisol. *Geoderma*. 312:6–16. doi: [10.1016/j.geoderma.2017.09.033](https://doi.org/10.1016/j.geoderma.2017.09.033).
- Deng J, Ma C, Zhang Y. 2022. Shallow landslide characteristics and its response to vegetation by example of July 2013, extreme rainstorm, central loess plateau, China. *Bull Eng Geol Environ*. 81(3):100. doi: [10.1007/s10064-022-02606-1](https://doi.org/10.1007/s10064-022-02606-1).
- Ding H, Xue L, Liu H, Li L, Wang H, Zhai M. 2022. Influence of root volume, plant spacing, and planting pattern of tap-like tree root system on slope protection effect. *Forests*. 13(11):1925–1925. doi: [10.3390/f13111925](https://doi.org/10.3390/f13111925).
- Greenway DR. 1987. Vegetation and slope stability In: Anderson MG, Richards KS. editors *Slope stability*. Chichester (UK): John Wiley & Sons; p. 187–230.
- El Kateb H, Zhang H, Zhang P, Mosandl R. 2013. Soil erosion and surface runoff on different vegetation covers and slope gradients: a field experiment in southern Shaanxi province, China. *Catena*. 105:1–10. doi: [10.1016/j.catena.2012.12.012](https://doi.org/10.1016/j.catena.2012.12.012).
- Eshel A, Beekman T. 2013. *Plant roots: the hidden half*. Boca Raton, FL: CRC Press LLC.
- Fan CC, Lu JZ, Chen HH. 2021. The pull-out resistance of plant roots in the field at different soil water conditions and root geometries. *Catena*. 207:105593. doi: [10.1016/j.catena.2021.105593](https://doi.org/10.1016/j.catena.2021.105593).
- Flepp G, Robyr R, Scotti R, Giadrossich F, Conedera M, Vacchiano G, Fischer C, Ammann P, May D, Schwarz M. 2021. Temporal dynamics of root reinforcement in European spruce forests. *Forests*. 12(6):815–815. doi: [10.3390/f12060815](https://doi.org/10.3390/f12060815).
- Francini A, Toscano S, Romano D, Ferrini F, Ferrante A. 2021. Biological contribution of ornamental plants for improving slope stability along urban and suburban areas. *Horticulturae*. 7(9):310. doi: [10.3390/horticulturae7090310](https://doi.org/10.3390/horticulturae7090310).
- Freschet GT, Pagès L, Iversen CM, Comas LH, Rewald B, Roumet C, Klimešová J, Zadworny M, Poorter H, Postma JA, Adams T S. et al. 2021. A starting guide to root ecology: strengthening ecological concepts and standardising root classification, sampling, processing and trait measurements. *New Phytol*. 232(3):973–1122. doi: [10.1111/nph.17572](https://doi.org/10.1111/nph.17572).
- Fu J, Xia Song Hu, Brierley G, Qiao N, Qin Qin Yu, Hai Jing Lu, Guo Rong Li, Zhu H. 2016. The influence of plant root system architectural properties upon the stability of loess hillslopes, Northeast Qinghai, China. *J Mt Sci*. 13(5):785–801. doi: [10.1007/s11629-014-3275-y](https://doi.org/10.1007/s11629-014-3275-y).
- Gao H, Xu X, Zhang H, Jiang Y, Zhao T. 2020. How effective is vegetation in reducing gravity erosion on loess gully sidewall under intense rainfalls? *Land Degrad Dev*. 31(17):2605–2619. doi: [10.1002/ldr.3634](https://doi.org/10.1002/ldr.3634).
- Genet M, Kokutse N, Stokes A, Fourcaud T, Cai X, Ji J, Mickovski S. 2008. Root reinforcement in plantations of *cryptomeria japonica* D. Don: effect of tree age and stand structure on slope stability. *For Ecol Manag*. 256(8):1517–1526. doi: [10.1016/j.foreco.2008.05.050](https://doi.org/10.1016/j.foreco.2008.05.050).
- Genet M, Stokes A, Salin F, Mickovski SB, Fourcaud T, Dumail J-F, van Beek R. 2005. The influence of cellulose content on tensile strength in tree roots. *Plant Soil*. 278(1–2):1–9. doi: [10.1007/s11104-005-8768-6](https://doi.org/10.1007/s11104-005-8768-6).
- Gerke KM, Sidle RC, Mallants D. 2015. Preferential flow mechanisms identified from staining experiments in forested hillslopes. *Hydrol Process*. 29(21):4562–4578.
- Ghestem M, Sidle RC, Stokes A. 2011. The influence of plant root systems on subsurface flow: implications for slope stability. *BioScience*. 61(11):869–879. doi: [10.1525/bio.2011.61.11.6](https://doi.org/10.1525/bio.2011.61.11.6).
- Gish TJ, Gimenez D, Rawls WJ. 1998. Impact of roots on ground water quality. *Plant and Soil*. 200:47–54. doi: [10.1023/A:1004202013082](https://doi.org/10.1023/A:1004202013082).
- Das Gk, Hazra B, Garg A, Wang C. 2018. Stochastic hydro-mechanical stability of vegetated slopes: an integrated copula based framework. *Catena*. 160:124–133. doi: [10.1016/j.catena.2017.09.009](https://doi.org/10.1016/j.catena.2017.09.009).
- Gomi T, Sidle RC, Miyata S, Kosugi K, Onda Y. 2008. Dynamic runoff connectivity of overland flow on steep forested hillslopes: scale effects and runoff transfer. *Water Resour Res*. 44(8). doi: [10.1029/2007WR005894](https://doi.org/10.1029/2007WR005894).
- Graham BF, Bormann FH. 1966. Natural root grafts. *Bot Rev*. 32 (3):255–292. doi: [10.1007/bf02858662](https://doi.org/10.1007/bf02858662).
- Gray DH, Leiser AT. 1982. *Biotechnical slope protection and erosion control*. New York: Van Nostrand Reinhold.
- Gray DH, Sotir RB. 1996. *Biotechnical and soil bioengineering slope stabilization: a practical guide for erosion control*. (New York): John Wiley & Sons.
- Hales TC, Miniati CF. 2017. Soil moisture causes dynamic adjustments to root reinforcement that reduce slope stability. *Earth Surf Process Landf*. 42(5):803–813.
- Hao G, Liu X, Li X. 2021. Quantification of additional cohesion from roots of mixed planting of multi-herb in dumping site and its influence on slope stability. *Arab J Geosci*. 14(8):739. doi: [10.1007/s12517-021-07089-0](https://doi.org/10.1007/s12517-021-07089-0).
- Hao G, Wang L, Liu X. 2023. Methods for studying the effect of plant roots on soil mechanical reinforcement: a review. *J Soil Sci Plant Nutr*. 23:2893–2912. doi: [10.1007/s42729-023-01330-3](https://doi.org/10.1007/s42729-023-01330-3).
- Hayati E, Abdi E, Saravi MM, Nieber JL, Majnounian B, Chirico GB, Wilson B, Nazarirad M. 2018. Soil water dynamics under different forest vegetation cover: implications for hillslope stability. *Earth Surf Process Landf*. 43(10):2106–2120. doi: [10.1002/esp.4376](https://doi.org/10.1002/esp.4376).
- Ide J, Shinohara Y, Higashi N, Komatsu H, Kuramoto K, Otsuki K. 2010. A preliminary investigation of surface runoff and soil properties in a moso-bamboo (*Phyllostachys pubescens*) forest in western Japan. *Hydrol Res Lett*. 4(2010):80–84. doi: [10.3178/HRL.4.80](https://doi.org/10.3178/HRL.4.80).
- Imaizumi, F., Sidle, R. C., & Kamei, R. 2008. Effects of forest harvesting on the occurrence of landslides and debris flows in steep terrain of central Japan. *Earth Surf Process Landf*. 33(6):827–840. doi: [10.1002/esp.1574](https://doi.org/10.1002/esp.1574).
- Iverson RM. 2015. Scaling and design of landslide and debris-flow experiments. *Geomorphology*. 244:9–20. doi: [10.1016/j.geomorph.2015.02.033](https://doi.org/10.1016/j.geomorph.2015.02.033).
- Jackson RB, Mooney HA, Schulze ED. 1997. A global budget for fine root biomass, surface area, and nutrient contents. *Proc Natl Acad Sci USA*. 94(14):7362–7366.
- Jarvis NJ. 2007. A review of non-equilibrium water flow and solute transport in soil macropores: principles, controlling factors and consequences for water quality. *Eur J Soil Sci*. 58(3):523–546. doi: [10.1111/j.1365-2389.2007.00915.x](https://doi.org/10.1111/j.1365-2389.2007.00915.x).
- Ji J, Mao Z, Qu W, Zhang Z. 2020. Energy-based fibre bundle model algorithms to predict soil reinforcement by roots. *Plant Soil*. 446 (1–2):307–329. doi: [10.1007/s11104-019-04327-z](https://doi.org/10.1007/s11104-019-04327-z).

- Jiang X, Liu W, Yang H, Li Z, Fan W, Wang F. 2023. A 3D model applied to analyze the mechanical characteristic of living stump slope with different tap root lengths. *Appl Sci*. 13(3):1978. doi: [10.3390/app13031978](https://doi.org/10.3390/app13031978).
- Johnsen K, Maier C, Kress L. 2005. Quantifying root lateral distribution and turnover using pine trees with a distinct stable carbon isotope signature. *Funct Ecol*. 19(1):81–87. doi: [10.1111/j.0269-8463.2005.00928.x](https://doi.org/10.1111/j.0269-8463.2005.00928.x).
- Johnson MS, Lehmann J. 2006. Double-funneling of trees: Stemflow and root-induced preferential flow. *Ecoscience*. 13(3):324–333. doi: [10.2980/11195-6860-13-3-324.1](https://doi.org/10.2980/11195-6860-13-3-324.1).
- Takeya R, Takizawa H, Kosaka I, Sonohara W, Ishigaki I, Abe K. 2016. Effects of thinning in *Cryptomeria japonica* D. Don stand on its root growth and function of preventing shallow landslide. *J Jpn Soc Reveget Technol*. 42(2):299–307. doi: [10.7211/jjsrt.42.299](https://doi.org/10.7211/jjsrt.42.299). (In Japanese with English abstract).
- Kamchoom V, Leung AK, Ng CW. 2014. Effects of root geometry and transpiration on pull-out resistance. *Géotechnique Lett*. 4(4):330–336.
- Kang Z, Lou G, Guo Y, Xu P. 2023. How does fine root influence aggregates? Comparing root diameter distribution of two herbaceous plants: a pot experiment. *J Soil Sci Plant Nutr*. 23(4):6377–6391. doi: [10.21203/rs.3.rs-2971159/v1](https://doi.org/10.21203/rs.3.rs-2971159/v1).
- Kessler M, Heller V, Turnbull B. 2018. A laboratory-numerical approach for modelling scale effects in dry granular slides. *Landslides*. 15:2145–2159. doi: [10.1007/s10346-018-1023-z](https://doi.org/10.1007/s10346-018-1023-z).
- Kinoshita A, Sakai Y, Tabata S, Ohno R, Kawashima M, Tamasaki T. 2013. Resistance force of laterally spread tree roots in Japanese cypress and cedar forests. *J Jpn Soc Eros Control Eng*. 65(5):1–20. doi: [10.11475/sabo.65.5_11](https://doi.org/10.11475/sabo.65.5_11). (In Japanese with English abstract).
- Kitahara H, Ono H, Atsui M. 2016. Distribution of the collapse prevention force of a root system in various forests. *Trans Jpn Geomorphol Union*. 37(4):455–463. (In Japanese with English abstract).
- Küllä T, Lohmus K. 1999. Influence of cultivation method on root grafting in Norway spruce (*Picea abies* (L.) karst.). *Plant Soil*. 217:91–100.
- Lan H, Wang D, He S, Fang Y, Chen W, Zhao P, Qi Y. 2020. Experimental study on the effects of tree planting on slope stability. *Landslides*. 17(4):1021–1035. doi: [10.1007/s10346-020-01348-z](https://doi.org/10.1007/s10346-020-01348-z).
- Lann T, Bao H, Lan H, Zheng H, Yan C. 2024. Hydro-mechanical effects of vegetation on slope stability: a review. *Sci Total Environ*. 926:171691. doi: [10.1016/j.scitotenv.2024.171691](https://doi.org/10.1016/j.scitotenv.2024.171691).
- Larcher W. 2003. *Physiological plant ecology ecophysiology and stress physiology of functional groups*. Berlin, Heidelberg: Springer.
- Leung AK, Garg A, Ng CW. 2015. Effects of plant roots on soil-water retention and induced suction in vegetated soil. *Eng Geol*. 193:183–197. doi: [10.1016/j.enggeo.2015.04.017](https://doi.org/10.1016/j.enggeo.2015.04.017).
- Leung AK, Kamchoom V, Ng Cww. 2017. Influences of root-induced soil suction and root geometry on slope stability: a centrifuge study. *Can Geotech J*. 54(3):291–303. doi: [10.1139/cgj-2015-0263](https://doi.org/10.1139/cgj-2015-0263).
- Li M, Ma C, Du C, Yang W, Lyu L, Wang X. 2021. Landslide response to vegetation by example of July 25–26, 2013, extreme rainstorm, tianshui, Gansu province, China. *Bull Eng Geol Environ*. 80:751–764. doi: [10.1007/s10064-020-02000-9](https://doi.org/10.1007/s10064-020-02000-9).
- Li R, Kan S, Zhu M, Chen J, Ai X, Chen Z, Zhang J, Ai Y. 2018. Effect of different vegetation restoration types on fundamental parameters, structural characteristics and the soil quality index of artificial soil. *Soil Tillage Res*. 184:11–23. doi: [10.1016/j.still.2018.06.010](https://doi.org/10.1016/j.still.2018.06.010).
- Liang T, Knappett JA. 2017a. Centrifuge modelling of the influence of slope height on the seismic performance of rooted slopes. *Géotechnique*. 67(10):855–869. doi: [10.1680/jgeot.16.p.072](https://doi.org/10.1680/jgeot.16.p.072).
- Liang T, Knappett JA. 2017b. Newmark sliding block model for predicting the seismic performance of vegetated slopes. *Soil Dyn Earthq Eng*. 101:27–40. doi: [10.1016/j.soildyn.2017.07.010](https://doi.org/10.1016/j.soildyn.2017.07.010).
- Liang T, Knappett JA, Duckett N. 2015. Modelling the seismic performance of rooted slopes from individual root–soil interaction to global slope behaviour. *Géotechnique*. 65(12):995–1009. doi: [10.1680/jgeot.14.P.207](https://doi.org/10.1680/jgeot.14.P.207).
- Liang T, Knappett JA, Leung AK, Carnaghan A, Bengough AG, Zhao R. 2019. A critical evaluation of predictive models for rooted soil strength with application to predicting the seismic deformation of rooted slopes. *Landslides*. 17(1):93–109. doi: [10.1007/s10346-019-01259-8](https://doi.org/10.1007/s10346-019-01259-8).
- Liang W-L, Kosugi K, Mizuyama T. 2011. Soil water dynamics around a tree on a hillslope with or without rainwater supplied by stemflow. *Water Resour Res*. 47(2):W02541. doi: [10.1029/2010WR009856](https://doi.org/10.1029/2010WR009856).
- Likitlersuang S, Takahashi A, Eab KH. 2017. Modeling of root-reinforced soil slope under rainfall condition. *Eng J*. 21(3):123–132. doi: [10.4186/ej.2017.21.3.123](https://doi.org/10.4186/ej.2017.21.3.123).
- Loades KW, Bengough AG, Bransby MF, Hallett PD. 2010. Planting density influence on fibrous root reinforcement of soils. *Eco Eng*. 36(3):276–284. doi: [10.1016/j.ecoleng.2009.02.005](https://doi.org/10.1016/j.ecoleng.2009.02.005).
- Löbmann MT, Geitner C, Wellstein C, Zerbe S. 2020. The influence of herbaceous vegetation on slope stability – a review. *Earth-Sci Rev*. 209:103328. doi: [10.1016/j.earscirev.2020.103328](https://doi.org/10.1016/j.earscirev.2020.103328).
- Lotfalian M, Nasiri M, Modarres A, Wu W. 2019. Slope stability analysis considering weight of trees and root reinforcement. *J Environ Eng Landsc Manag*. 27(4):201–208. doi: [10.3846/jeelm.2019.11292](https://doi.org/10.3846/jeelm.2019.11292).
- Lynch J. 1995. Root architecture and plant productivity. *Plant Physiol*. 109(1):7–13. doi: [10.1104/pp.109.1.7](https://doi.org/10.1104/pp.109.1.7).
- Makoto K, Kitagawa R, Blume-Werry G. 2023. How do leaf functional traits and age influence the maximum rooting depth of trees? *Eur J For Res*. 142(5):1197–1206. doi: [10.1007/s10342-023-01585-6](https://doi.org/10.1007/s10342-023-01585-6).
- Mao Z, Bourrier F, Stokes A, Fourcaud T. 2014. Three-dimensional modelling of slope stability in heterogeneous montane forest ecosystems. *Ecol Model*. 273:11–22. doi: [10.1016/j.ecolmodel.2013.10.017](https://doi.org/10.1016/j.ecolmodel.2013.10.017).
- Masi EB, Segoni S, Tofani V. 2021. Root reinforcement in slope stability models: a review. *Geosciences*. 11(5):212. doi: [10.3390/geosciences11050212](https://doi.org/10.3390/geosciences11050212).
- Mattia C, Bischetti GB, Gentile F. 2005. Biotechnical characteristics of root systems of typical Mediterranean species. *Plant Soil*. 278(1–2):23–32. doi: [10.1007/s11104-005-7930-5](https://doi.org/10.1007/s11104-005-7930-5).
- McGuire LA, Rengers FK, Kean JW, Jeffrey CA, Mirus BB, Baum RL, Godt JW. 2016. Elucidating the role of vegetation in the initiation of rainfall-induced shallow landslides: insights from an extreme rainfall event in the Colorado front range. *Landslides*. 13(5):1223–1237. doi: [10.1002/2016gl070741](https://doi.org/10.1002/2016gl070741).
- Mehtab A, Jiang YJ, Su LJ, Shamsher S, Li JJ, Mahfuzur R. 2021. Scaling the roots mechanical reinforcement in plantation of Cunninghamia R. Br in Southwest China For. 12(1):33. doi: [10.3390/f12010033](https://doi.org/10.3390/f12010033).
- Meng S, Zhao G, Yang Y. 2021. Experimental study on influence of vegetation roots on hydraulic characteristics of slope under rainfall. *J Phys Conf Ser*. 1838(1):012057. doi: [10.1088/1742-6596/1838/1/012057](https://doi.org/10.1088/1742-6596/1838/1/012057).
- Mickovski SB, van Beek, LPH. 2009. Root morphology and effects on soil reinforcement and slope stability of young vetiver (*Vetiveria zizanioides*) plants grown in semi-arid climate. *Plant Soil*. 324(1–2):43–56. doi: [10.1007/s11104-009-0130-y](https://doi.org/10.1007/s11104-009-0130-y).
- Milledge DG, Bellugi D, McKean JA, Densmore AL, Dietrich WE. 2014. A multidimensional stability model for predicting shallow landslide size and shape across landscapes. *J Geophys Res Earth Surf*. 119(11):2481–2504. doi: [10.1002/2014jf003135](https://doi.org/10.1002/2014jf003135).
- Mitchell AR, Ellsworth TR, Meek BD. 1995. Effect of root systems on preferential flow in swelling soil. *Commun Soil Sci Plant Anal*. 26(15–16):2655–2666. doi: [10.1080/00103629509369475](https://doi.org/10.1080/00103629509369475).
- Moos C, Bebi P, Graf F, Mattli J, Rickli C, Schwarz M. 2016. How does forest structure affect root reinforcement and susceptibility to shallow landslides? *Earth Surf Process Landf*. 41(7):951–960. doi: [10.1002/esp.3887](https://doi.org/10.1002/esp.3887).
- Ng CWW, Zhang Q, Ni J, Li Z. 2021. A new three-dimensional theoretical model for analysing the stability of vegetated slopes with different root architectures and planting patterns. *Comput Geotech*. 130:103912. doi: [10.1016/j.compgeo.2020.103912](https://doi.org/10.1016/j.compgeo.2020.103912).
- Ni JJ, Leung AK, Ng Cww. 2019. Influences of plant spacing on root tensile strength of *Schefflera arboricola* and soil shear strength. *Landsc Ecol Eng*. 15(2):223–230. doi: [10.1007/s11355-019-00374-x](https://doi.org/10.1007/s11355-019-00374-x).
- Nicoll BC, Ray D. 1996. Adaptive growth of tree root systems in response to wind action and site conditions. *Tree Physiol*. 16(11–12):891–898. doi: [10.1093/treephys/16.11-12.891](https://doi.org/10.1093/treephys/16.11-12.891).
- Noguchi S, Tsuboyama Y, Sidle RC, Hosoda I. 1999. Morphological characteristics of macropores and the distribution of preferential flow pathways in a forested slope segment. *Soil Sci Soc Am J*. 63(5):1413–1423. doi: [10.2136/sssaj1999.6351413x](https://doi.org/10.2136/sssaj1999.6351413x).
- Norris JE, Stokes A, Mickovski SB, Cammeraat E, van Beek LPH, Nicoll B, Achim A, editors. 2008. *Slope stability and erosion control: ecotechnological solutions*. Dordrecht (Netherlands): Springer. 10.1007/978-1-4020-6676-4.

- Noviandi R, Gomi T, Kharismalatri HS, Sidle RC, Ritonga RP, Shiraki K. 2022. The mobility of landslides in pumice: insights from a flume experiment. *Water*. 14(19):3083. doi: [10.3390/w14193083](https://doi.org/10.3390/w14193083).
- Ochiai H, Sammori T, Okada Y. 2007. Landslide experiments on artificial and natural slopes. In: Sassa K, Fukuoka H, Wang F, Wang G., edited by. *Progress in landslide science*. 209–226.
- Oorthuis R, Hürlimann M, Fraccica A, Lloret A, Moya J, Puig-Polo C, Vaunat J. 2018. Monitoring of a full-scale embankment experiment regarding soil-vegetation-atmosphere interactions. *Water*. 10(6):688. doi: [10.3390/w10060688](https://doi.org/10.3390/w10060688).
- Osman N, Barakbah SS. 2011. The effect of plant succession on slope stability. *Eco Eng*. 37(2):139–147. doi: [10.1016/j.ecoleng.2010.08.002](https://doi.org/10.1016/j.ecoleng.2010.08.002).
- Pollen N. 2007. Temporal and spatial variability in root reinforcement of streambanks: accounting for soil shear strength and moisture. *Catena*. 69(3):197–205. doi: [10.1016/j.catena.2006.05.004](https://doi.org/10.1016/j.catena.2006.05.004).
- Pollen N, Simon A. 2005. Estimating the mechanical effects of riparian vegetation on stream bank stability using a fiber bundle model. *Water Resour Res*. 41(7):W07025. doi: [10.1029/2004WR003801](https://doi.org/10.1029/2004WR003801).
- Raats PAC. 2007. Uptake of water from soils by plant roots. *Transp Porous Media*. 68(1):5–28. doi: [10.1007/s11242-006-9055-6](https://doi.org/10.1007/s11242-006-9055-6).
- Ranjan V, Sen P, Kumar D, Saraswat A. 2016. Enhancement of mechanical stability of waste dump slope through establishing vegetation in a surface iron ore mine. *Environ Earth Sci*. 76(1):35. doi: [10.1007/s12665-016-6350-6](https://doi.org/10.1007/s12665-016-6350-6).
- Ravenek JM, Bessler H, Engels C, Scherer-Lorenzen M, Gessler A, Gockele A, De Luca E, Temperton VM, Ebeling A, Roscher C, Schmid B, Weisser WB, Wirth C, de Kroon H, Weigelt A, Mommer, L. 2014. Long-term study of root biomass in a biodiversity experiment reveals shifts in diversity effects over time. *Oikos*. 123(12):1528–1536. doi: [10.1111/oik.01502](https://doi.org/10.1111/oik.01502).
- Rickli C, Graf F. 2009. Effects of forests on shallow landslides – case studies in Switzerland. *For Snow Landsc Res*. 82:33–44.
- Ritonga RP, Gomi T, Tarigan SD, Kaswanto RL, Kharismalatri HS, Noviandi R, Arata Y, Ishikawa Y. 2021. Land cover and characteristics of landslides induced by the 2018 MW 6.7 eastern Iburi earthquake, Hokkaido. *Int J Eros Control Eng*. 13(4):76–83. doi: [10.13101/ijece.13.76](https://doi.org/10.13101/ijece.13.76).
- Roering JJ, Schmidt KM, Stock JD, Dietrich WE, Montgomery DR. 2003. Shallow landsliding, root reinforcement, and the spatial distribution of trees in the Oregon Coast range. *Can Geotech J*. 40(2):237–253. doi: [10.1139/t02-113](https://doi.org/10.1139/t02-113).
- Rui D, Ji M, Nakamur D, Suzuki T. 2018. Experimental study on gravitational erosion process of vegetation slope under freeze-thaw. *Cold Reg Sci Technol*. 151:168–178. doi: [10.1016/j.coldreg.2018.03.020](https://doi.org/10.1016/j.coldreg.2018.03.020).
- Sakals ME, Sidle RC. 2004. A spatial and temporal model of root cohesion in forest soils. *Can J For Res*. 34(4):950–958. doi: [10.1139/x03-268S](https://doi.org/10.1139/x03-268S).
- Sato H, Otani K, Kanbara T, Torita H. 2009. The effect of root system in cool-temperate deciduous broad-leaved forest on the slope stability. *J Jpn Soc Eros Control Eng*. 62(4):29–37. doi: [10.11475/sabo.62.4_29](https://doi.org/10.11475/sabo.62.4_29). (In Japanese with English abstract).
- Sato H, Torita H, Masaka K. 2001. Root growth of todo fir after thinning. *Trans Meet Hokkaido Branch Jpn For Soc*. 49:118–119. doi: [10.24494/jfshb.49.0_118](https://doi.org/10.24494/jfshb.49.0_118). (In Japanese).
- Schenk HJ. 2006. Root competition: beyond resource depletion. *J Ecol*. 94(4):725–739. doi: [10.1111/j.1365-2745.2006.01124.x](https://doi.org/10.1111/j.1365-2745.2006.01124.x).
- Schenk HJ, Jackson RB. 2002. The global biogeography of roots. *Ecol Monogr*. 72(3):311–328. doi: [10.2307/3100092](https://doi.org/10.2307/3100092).
- Schmidt KM, Roering JJ, Stock JD, Dietrich WE, Montgomery DR, Schaub T. 2001. The variability of root cohesion as an influence on shallow landslide susceptibility in the Oregon Coast range. *Can Geotech J*. 38(5):995–1024. doi: [10.1139/t01-031](https://doi.org/10.1139/t01-031).
- Schwarz M, Cohen D, Or D. 2011. Pull-out tests of root analogs and natural root bundles in soil: experiments and modeling. *J Geophys Res Earth Surf*. 116:F02007. doi: [10.1029/2010JF001753](https://doi.org/10.1029/2010JF001753).
- Schwarz M, Cohen D, Or D. 2012. Spatial characterization of root reinforcement at stand scale: theory and case study. *Geomorphology*. 171–172:190–200. doi: [10.1016/j.geomorph.2012.05.020](https://doi.org/10.1016/j.geomorph.2012.05.020).
- Schwarz M, Giadrossich F, Cohen D. 2013. Modeling root reinforcement using a root-failure Weibull survival function. *Hydrol Earth System Sci*. 17(11):4367–4377. doi: [10.5194/hess-17-4367-2013](https://doi.org/10.5194/hess-17-4367-2013).
- Schwarz M, Lehmann P, Or D. 2010. Quantifying lateral root reinforcement in steep slopes - from a bundle of roots to tree stands. *Earth Surf Process Landf*. 35(3):354–367. doi: [10.1002/esp.1927](https://doi.org/10.1002/esp.1927).
- Schwarz M, Phillips C, Marden M, McIvor Ir, Douglas GB, Watson A. 2016. Modelling of root reinforcement and erosion control by ‘veronese’ Poplar on pastoral hill country in New Zealand. *N Z J For Sci*. 46(1):4. doi: [10.1186/s40490-016-0060-4](https://doi.org/10.1186/s40490-016-0060-4).
- Shao W, Ni J, Leung AK, Su Y, Ng Cww. 2017. Analysis of plant root-induced preferential flow and pore-water pressure variation by a dual-permeability model. *Can Geotech J*. 54(11):1537–1552. doi: [10.1139/cgj-2016-0629](https://doi.org/10.1139/cgj-2016-0629).
- Shi X, Qin T, Yan D, Tian F, Wang H. 2021. A meta-analysis on effects of root development on soil hydraulic properties. *Geoderma*. 403:115363. doi: [10.1016/j.geoderma.2021.115363](https://doi.org/10.1016/j.geoderma.2021.115363).
- Shuin Y, Kato H, Suzuki M, Ohta T. 1998. Effects of soil moisture fluctuation on soil reinforcement by root -direct shear tests with materials simulating a root system-. *J Jpn Soc Eros Control Eng*. 51(1):23–30. doi: [10.11475/sabo1973.51.23](https://doi.org/10.11475/sabo1973.51.23).
- Sidle RC. 1992. A theoretical model of the effects of timber harvesting on slope stability. *Water Resour Res*. 28(7):1897–1910. doi: [10.1029/92wr00804](https://doi.org/10.1029/92wr00804).
- Sidle RC, Gomi T, Loaiza Usuga JC, Jarihani B. 2017. Hydrogeomorphic processes and scaling issues in the continuum from soil pedons to catchments. *Earth-Sci Rev*. 175:75–96. doi: [10.1016/j.earscirev.2017.10.010](https://doi.org/10.1016/j.earscirev.2017.10.010).
- Sidle RC, Noguchi S, Tsuboyama Y, Laursen K. 2001. A conceptual model of preferential flow systems in forested hillslopes: evidence of self-organization. *Hydrol Process*. 15(10):1675–1692. doi: [10.1002/hyp.233](https://doi.org/10.1002/hyp.233).
- Sidle RC, Ochiai H. 2006. Landslides: processes, prediction, and land use. (Washington) (DC): American Geophysical Union. [10.1029/wm018](https://doi.org/10.1029/wm018).
- Stokes A, Atger C, Bengough AG, Fourcaud T, Sidle RC. 2009. Desirable plant root traits for protecting natural and engineered slopes against landslides. *Plant Soil*. 324(1–2):1–30. doi: [10.1007/s11104-009-0159-y](https://doi.org/10.1007/s11104-009-0159-y).
- Takahashi T, Awaya Y, Hirata Y, Furuya N, Sakai T, Sakai A. 2010. Stand volume estimation by combining low laser-sampling density LiDAR data with QuickBird panchromatic imagery in closed-canopy Japanese cedar (*Cryptomeria japonica*) plantations. *Int J Remote Sens*. 31(5):1281–1301. doi: [10.1080/01431160903380623](https://doi.org/10.1080/01431160903380623).
- Tan X, Huang Y, Xiong D, Lv K, Chen F. 2020. The effect of Elymus nutans sowing density on soil reinforcement and slope stabilization properties of vegetation-concrete structures. *Sci Rep*. 10(1):20462. doi: [10.1038/s41598-020-77407-1](https://doi.org/10.1038/s41598-020-77407-1).
- Tarroux E, DesRochers A, Krause C. 2010. Effect of natural root grafting on growth response of jack pine (*Pinus banksiana*) after commercial thinning. *For Ecol Manag*. 260(4):526–535. doi: [10.1016/j.foreco.2010.05.008](https://doi.org/10.1016/j.foreco.2010.05.008).
- Tomobe H, Tsugawa S, Yoshida Y, Yamada M. 2023. A mechanical theory of competition between plant root growth and soil pressure reveals a potential mechanism of root penetration. *Sci Rep*. 13(1):7473. doi: [10.1038/s41598-023-34025-x](https://doi.org/10.1038/s41598-023-34025-x).
- Tsukamoto Y, Kubota J. 1991. Roles of forest in hillslope stability. *Trans Jpn Geomorphol Union*. 12(3):243–257. (In Japanese with English abstract).
- Tsukamoto Y, Minematsu H, Fujinami T. 1986. Effect of tree roots on slope stability: mainly on the effect of lateral roots. *J Jpn Assoc Reveget Technol*. 12(1):11–20. doi: [10.7211/jjsrt1973.12.11](https://doi.org/10.7211/jjsrt1973.12.11). (In Japanese).
- Tsukamoto Y, Ohta T, Kitahara I, Tsuru H. 1982. Simulation model of tree-root distributions of a stand. *J Jpn For Soc*. 64(2):56–65. doi: [10.11519/jjfs1953.72.5_375](https://doi.org/10.11519/jjfs1953.72.5_375). (In Japanese with English abstract).
- Tsutsumi D, Kosugi K, Mizuyama T. 2004. Three-dimensional modeling of hydrotropism effects on plant root architecture along a hillslope. *Vadose Zone J*. 3(3):1017–1030. doi: [10.2136/vzj2004.1017](https://doi.org/10.2136/vzj2004.1017).
- Waldron LJ. 1977. The shear resistance of root-permeated homogeneous and stratified soil. *Soil Sci Soc Am J*. 41(5):843–849. doi: [10.2136/sssaj1977.03615995004100050005x](https://doi.org/10.2136/sssaj1977.03615995004100050005x).
- Wang L, Zhang J, Zhao Y, Fu Q, Li T. 2022. Vegetation restoration and plant roots improve soil infiltration capacity after a severe forest fire in Daxing'anling, northeast China. *J Soil Water Conserv*. 77(2):135–143. doi: [10.2489/jswc.2022.00059](https://doi.org/10.2489/jswc.2022.00059).
- Wang S, Zhao M, Meng X, Chen G, Zeng R, Yang Q, Liu Y, Wang B. 2020. Evaluation of the effects of forest on slope stability and its

- implications for forest management: a case study of Bailong River Basin, China. *Sustainability*. 12(16):6655. doi: [10.3390/su12166655](https://doi.org/10.3390/su12166655).
- Wang X, Huang Z, Hong MM, Zhao YF, Ou YS, Zhang J. 2019. A comparison of the effects of natural vegetation regrowth with a plantation scheme on soil structure in a geological hazard-prone region. *Eur J Soil Sci*. 70(3):674–685. doi: [10.1111/ejss.12781](https://doi.org/10.1111/ejss.12781).
- Wang X, Ma C, Wang Y, Wang Y, Li T, Dai Z, Li M. 2020. Effect of root architecture on rainfall threshold for slope stability: variabilities in saturated hydraulic conductivity and strength of root-soil composite. *Landslides*. 17(8):1965–1977. doi: [10.1007/s10346-020-01422-6](https://doi.org/10.1007/s10346-020-01422-6).
- Wang X-P, Li X-R, Xiao H-L, Berndtsson R, Pan Y-X. 2007. Effects of surface characteristics on infiltration patterns in an arid shrub desert. *Hydrol Process*. 21(1):72–79. doi: [10.1002/hyp.6185](https://doi.org/10.1002/hyp.6185).
- Wu TH, Beal PE, Lan C. 1988. In-situ shear test of soil-root systems. *J Geotech Eng*. 114(12):1376–1394. doi: [10.1061/\(asce\)0733-9410\(1988\)114:12\(1376\)](https://doi.org/10.1061/(asce)0733-9410(1988)114:12(1376)).
- Wu TH, McKinnell III WP, Swanston DN. 1979. Strength of tree roots and landslides on Prince of Wales Island, Alaska. *Can Geotech J*. 16(1):19–33. doi: [10.1139/t79-003](https://doi.org/10.1139/t79-003).
- Yamase K, Ikeno H, Hotta N, Imawaka M, Ohashi M, Tanikawa T, Todo C, Dannoura M, Hirano Y. 2024. Effect of sprouting and corresponding root distribution of the shrub species *Eurya japonica* on slope stability. *Catena*. 238:107869. doi: [10.1016/j.catena.2024.107869](https://doi.org/10.1016/j.catena.2024.107869).
- Yamase K, Tanikawa T, Dannoura M, Ohashi M, Todo C, Ikeno H, Aono K, Hirano Y. 2018. Ground-penetrating radar estimates of tree root diameter and distribution under field conditions. *Trees (Berl)*. 32(6):1657–1668. doi: [10.1007/s00468-018-1741-9](https://doi.org/10.1007/s00468-018-1741-9).
- Yamase K, Tanikawa T, Dannoura M, Todo C, Yamamoto T, Ikeno H, Ohashi M, Aono K, Doi R, Hirano Y. 2019. Estimating slope stability by lateral root reinforcement in thinned and unthinned stands of *cryptomeria japonica* using ground-penetrating radar. *Catena*. 183:104227. doi: [10.1016/j.catena.2019.104227](https://doi.org/10.1016/j.catena.2019.104227).
- Yamase K, Todo C, Torii N, Tanikawa T, Yamamoto T, Ikeno H, Ohashi M, Dannoura M, Hirano Y. 2021. Dynamics of soil reinforcement by roots in a regenerating coppice stand of *Quercus serrata* and effects on slope stability. *Eco Eng*. 162:106169. doi: [10.1016/j.ecoleng.2021.106169](https://doi.org/10.1016/j.ecoleng.2021.106169).
- Yanagisawa N, Fujita N. 1999. Different distribution patterns of woody species on a slope in relation to vertical root distribution and dynamics of soil moisture profiles. *Ecol Res*. 14(2):165–177. doi: [10.1046/j.1440-1703.1999.00295.x](https://doi.org/10.1046/j.1440-1703.1999.00295.x).
- Yildiz A, Graf F, Rickli C, Springman SM. 2018. Determination of the shearing behaviour of root-permeated soils with a large-scale direct shear apparatus. *Catena*. 166:98–113. doi: [10.1016/j.catena.2018.03.022](https://doi.org/10.1016/j.catena.2018.03.022).
- Zhang C, Liu Y, Li D, Jiang J. 2020. Influence of soil moisture content on pull-out properties of *Hippophae rhamnoides* Linn. roots. *J Mt Sci*. 17(11):2816–2826. doi: [10.1007/s11629-020-6072-9](https://doi.org/10.1007/s11629-020-6072-9).
- Zhang X, Yu GQ, Li ZB, Li P. 2014. Experimental study on slope runoff, erosion and sediment under different vegetation types. *Water Resour Manage*. 28(9):2415–2433. doi: [10.1007/s11269-014-0603-5](https://doi.org/10.1007/s11269-014-0603-5).
- Ziemer RR. 1981. Roots and the stability of forested slopes. In: Davies TRH, Pearce AJ editors. *Proceedings of the International Symposium on Erosion and Sediment Transport in Pacific Rim Steeplands*; [1981 Jan 25–31]; Christchurch, NZ. Wallingford (UK): International Association of Hydrological Sciences. p. 343–361. (IAHS Publication No. 132).